What Makes an Opera a Staple of the Repertoire?

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1. Introduction

Although the 19th century witnessed a surge in lyrical production that led to the formation of an operatic canon, only a handful of works have endured over time; this canon, torn between a reverence for classics and desire for innovation, is the result of a process of social and institutional selection (*e.g.* the stabilisation of the operatic curation) (Charle, 2021). Examining the repertoire of works from the Opéra de Paris could thus shed light on the mechanisms through which this canon was shaped. But we also sought in our analysis to incorporate a cultural dimension, by studying whether the presence of an innovative element for the time of the composition of an opera could have an impact on its success today (*ibid.*); we also introduced in the operatic repertoire the notion of aesthetic gap theorised by Jauss in literature, which corresponds to the gap between a work and the norms of an era (Jauss, 1978). Finally, we aimed to replicate a previous analysis conducted by PSL students, but using different methods and considering the canonical status of a work instead of its number of representations nowadays. To do so, we computed our measure of aesthetic gap, canonisation and presence of innovative element for each individual work from the repertoire using a large language model.

Does the aesthetic gap of a lyrical work have an influence on its being considered canonical in the repertoire of a given period?

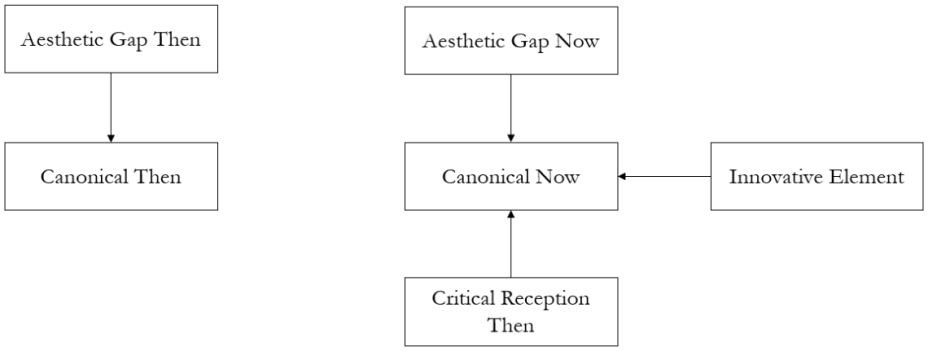


Fig. 1: Directed acyclic graph representing the hypothesised links between: to the left, aesthetic gap with respect to the standard of its composition time and its belonging to the canon of this period; to the right, the aesthetic gap with today's standard, its belonging to the present canon, and the presence of an innovative element.

We anticipate that if this is the case, we should observe:

- **P1 –** For a given opera, the probability of belonging to the canon of its composition time should decrease as the aesthetic gap with its composition time increases.
- **P2 –** For a given opera, the probability of belonging to today's canon should decrease as the aesthetic gap with respect to nowadays standard repertoire increases…
- **P3 –** … and that it should increase if it is more likely to have contained a groundbreaking innovative element for its composition time.
- **P4 –** For a given opera, the probability of belonging to today's canon should increase as the critical reception at the time is better.

2. Data

2.1. Collection Methodology

The dataset we used, MarisaFerrer5, is an extended version of the MarisaFerrer3 dataset, created in spring 2024 by Sacha Levi Mazloum. It initially gathered data about 479 lyrical works premiered in the Opéra de Paris between 1749 and 1989, providing general information about their composers, styles, librettists, but also the possibility to study the evolution of repertoire with quantitative data about the number of representations ten years after the premiere, as well as in the last 30 years. An attempt to complete the dataset with more qualitative data had been made with the addition of the "critical reception" column, after numerous and tedious hours of manual annotation of press articles on Gallica.

To extend the dataset further with information relevant to our analysis, Éloïse Delerue conducted the automation of the annotation, using the GPT-4 language model developed by OpenAI and following this methodology (Dubourg, Thouzeau & Baumard, 2024). We first performed prompt engineering, running several tests on a small amount of data from MarisaFerrer3 until the model was sufficiently guided to provide satisfactory answers. Throughout the process, we performed external validity checks, verifying the information through Sacha Levi Mazloum's expertise, GPT-4's justification, and comparison with other pre-existing columns in the dataset.

Finally, MarisaFerrer5, created in spring 2025, in addition to providing features from MarisaFerrer3, also provides information on the 479 works's inspiration, themes, aesthetic gap with the period of creation and our contemporary period, presence of an innovative element (if yes, which one), and position in the canon of the period of creation and our contemporary period, as well as the model's justification (see **Fig. 2**, **Fig. 3** and **Fig. 4** below).

Before venturing on the analysis, we took a quick look at each model justification and removed from the dataset works for which they seemed too obviously uncertain or ignorant (tracking for use of modals, avowals of lack of knowledge, mistakes in the composer's name) (for details of the annotation, see the attached notebook **MarisaFerrer5_Annotation_Delerue_Levi**).

2.2. Description

2.2.1. Monovariate Description

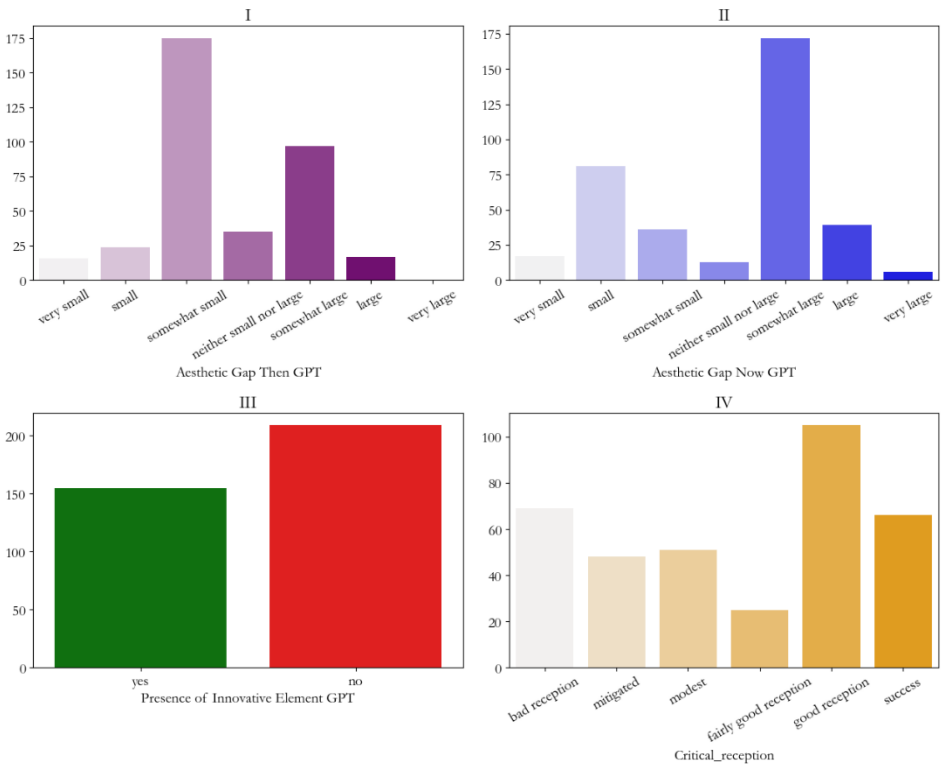
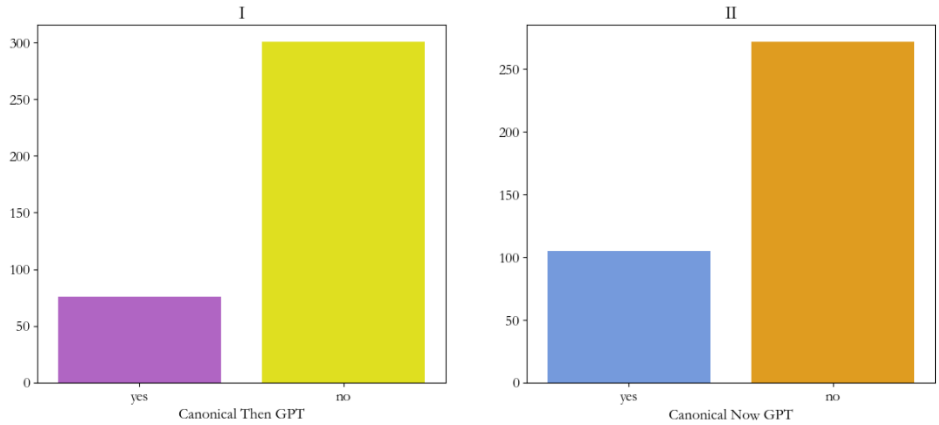


Fig. 2: Distribution of our predictors. The annotated aesthetic gap (**I**) at the time the opera was composed and (**II**) with respect to today's standards. (**III**) The presence of a groundbreaking innovative element for the time. (**IV**) The critical reception at the time.

Fig. 3: Distribution of the canonical status. (I) At the time the opera was composed and **(II)** nowadays.



2.2.1. Bivariate Description

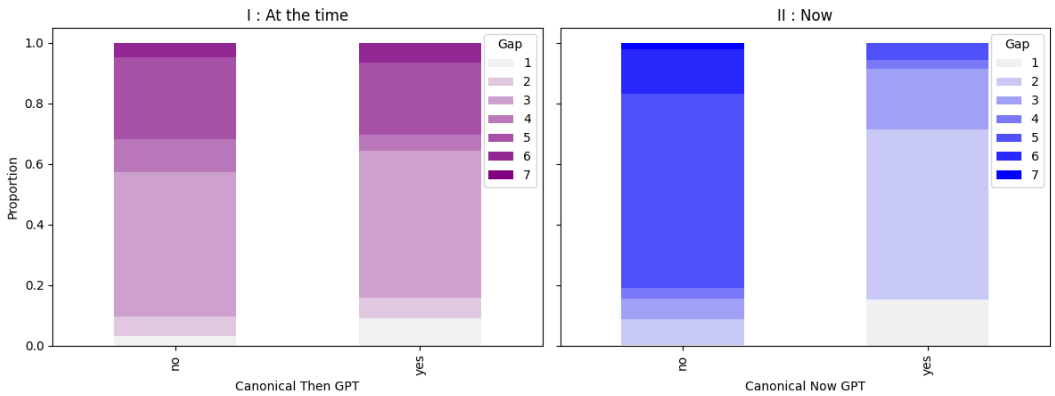


Fig. 4: Distribution of aesthetic gap depending on the canonical status. As proportions, **(I)** at the time the opera was composed and **(II)** nowadays.

3. Methods

To test our predictions, we performed two Bayesian logistic regression using the PyMc library. In these probabilistic models, a so-called "posterior" probability distribution of a regression parameter ($\beta 1$, $\beta 2$, etc.), knowing the data, is estimated running a MCMC (Markov Chain Monte Carlo) sampling algorithm. For **P1**, the probabilistic **model I** corresponds to the following formula[1]:

$$Canonical_then_i \sim Bernoulli(p_i)\ with\ logit(p_i) = \beta_0 + \beta_1 Aesthetic_gap_then_i + \epsilon_i$$

with:

- **Canonical_then$_i$:** Canonical status of the opera i at the time (binary variable, see **Fig. 2, I**);
- **Aesthetic_gap_then$_i$:** Aesthetic gap between the opera i and the standards of its time (ordinal variable, ranging from 1 to 7 see **Fig. 3, I**).

To test **P2**, **P3** and **P4**, our **model II** is built in the same fashion:

$$\begin{aligned} Canonical_now_i \sim Bernoulli(p_i)\ with\ logit(p_i) \\ = \beta_0 + \beta_1 Aesthetic_gap_now_i \\ + \beta_2 Innovative_element_i + \beta_3 Critical_reception_i + \epsilon_i \end{aligned}$$

[1] Even though our variables are not standardized, we shall henceforth notate β_i the regression coefficients, in accordance with Bayesian conventions.

with:

- **Canonical_now$_i$:** Canonical status of the opera i nowadays (binary variable, see **Fig. 1, II**);
- **Aesthetic_gap_now$_i$:** Aesthetic gap between the opera i and the standards opera repertoire of today (ordinal variable, ranging from 1 to 7 see **Fig. 3, II**);
- **Innovative_element$_i$:** presence of a groundbreaking innovative element in its time for opera i (binary variable, see **Fig. 2, III**).
- **Critical_reception$_i$:** indicates how well was the work i received in its time by critics, on a scale from 1 to 6 (ordinal variable, see **Fig. 2, IV**).

We assigned priors on regression coefficients so that they are in line with our predictions, but without making them too strong (β_1 normally distributed around -0.5 for **models I** and **II**; β_2 normally distributed around +0.5) (for details of the analysis, see the attached notebook **MarisaFerrer5_Analysis_Levi_Delerue**).

4. Results

Our first test results suggest a slight negative association between the aesthetic gap with respect to the standards of the time, and the probability of having been considered canonical then (posterior distribution for β_1 of mean -0.13), the uncertainty around this association given the data is too high to conclude (see **Table 1**).

Predictor	**posterior mean**	**posterior sd**	**HDI 94%**
Aesthetic Gap Then	-0.098	0.107	[-0.3,0.095]

Table 1: Result of Bayesian logistic regression predicting Canonical Then.

As for the second test, its results suggest a robust negative association between the aesthetic gap with respect to today's standard repertoire and the probability for a lyrical work of being considered canonical today (see **Table 2**).

Predictor	**posterior mean**	**posterior sd**	**HDI 94%**
Aesthetic Gap Now	-1.4	0.161	[-1.7,-1.1]
Innovative Element	1.3	0.397	[0.51,2]
Critical Reception	-0.23	0.105	[-0.42,-0.038]

Table 2: Result of Bayesian logistic regression predicting Canonical Now.

5. Discussion

Given our data, a robust negative association between the aesthetic gap with respect to today's standard repertoire and the probability for a lyrical work of being considered canonical today is

plausible with a high degree of certainty. These results may seem obvious, but we also report an interesting robust positive association between the presence of an innovative element for the time, and the probability of having achieved canonical status nowadays. Thus, **we can confidently confirm P2 and P3**.

However, given our data, an association between aesthetic gap and canonical status at the time is too uncertain for us to conclude, even with the prior intuition that it should be a negative one. We can observe that the comparative stacked bar charts (see **Fig. 4**) of aesthetic gap and canonic status now and then, already gave us a hint of this discrepancy in the results. Thus, **we cannot confirm P1**.

At last we ought to announce baffling results concerning the replication of the positive association between critical reception (at the time) and popularity today: with the now canonical status as dependent variable and a logistic bayesian regression model, the association appears to be positive and quite robust. Thus, **we can invalidate P4**.

Please note that we used the PyMc sampling method as it is widely used, but that Bayesian results depend on the methods. We did not sample much either because we had only little data. Furthermore, despite our best efforts, the possibility of haphazard annotation for little-known operas should be considered.

We found again a positive robust association with critical reception when this was used alone as a predictor; interestingly its effect no longer seems to hold when controlled for the effect of other variables. Therefore, we may wonder why the association between critical reception at the time of the creation of the opera and its popularity today is positive. Another future analysis might be to actually test Charle's thesis that today's canon has its roots in the canon formed from the 19th century onwards.

6. References

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Dubourg, E., Thouzeau, V., & Baumard, N. (2024). A step-by-step method for cultural annotation by LLMs. Frontiers in Artificial Intelligence, 7. https://doi.org/10.3389/frai.2024.1365508

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